IIT Bombay Driverless Recruitment Hackathon Report

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1 Perception

The perception module is responsible for detecting cones in the environment and estimating their spatial positions. A YOLO-based detection model is used to identify cones in the camera image. The model outputs bounding boxes around detected cones, from which the height of the cone in pixels can be obtained.

1.1 Depth Estimation

To estimate the distance of the cone from the camera, the pinhole camera model is used. This model is based on the similarity of triangles formed by the object, the camera, and the image plane.

$$Z = \frac{fH}{h} \tag{1}$$

Where:

- Z = distance of the cone from the camera
- f = focal length of the camera
- H = real-world height of the cone
- h = height of the cone in pixels in the image

The OpenCV library is used to visualize detections by drawing bounding boxes and labels on the image.

1.2 Assumptions

- The detection model provides accurate bounding box coordinates.
- The real-world height of cones is constant.
- The camera focal length is known and fixed.

2 Planning and Control (PPC)

2.1 Planner

The planner constructs a drivable path from the detected cone positions.

Each yellow cone is paired with the nearest blue cone. The midpoint between each pair is computed and used as a waypoint representing the center of the track.

$$(x_m, y_m) = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right) \tag{2}$$

Where (x_1, y_1) and (x_2, y_2) represent the positions of the paired cones.

These midpoints represent the centerline of the track. Additional interpolation is performed to create a smoother trajectory for the vehicle.

2.2 Controller

The controller determines steering and throttle commands to follow the planned trajectory.

2.2.1 Steering

For steering control, the nearest waypoint to the current vehicle position is first identified. A lookahead index is added to this waypoint (for example +10) to produce smoother steering behavior.

The steering command is calculated using the difference between the heading of the vehicle and the direction toward the selected waypoint.

$$\theta_{error} = \arctan\left(\frac{y_{target} - y_{current}}{x_{target} - x_{current}}\right) - \psi \tag{3}$$

Where:

- ψ is the current yaw of the vehicle
- (x_{target}, y_{target}) is the selected waypoint

2.2.2 Throttle Control

The throttle controller is implemented using a proportional control strategy.

$$e_v = v_{target} - v_{current} \tag{4}$$

If the error is positive, throttle is applied proportionally to accelerate the vehicle. If the error is negative, braking is applied instead.

The magnitude of the velocity vector is used instead of only the x-component:

$$|v| = \sqrt{v_x^2 + v_y^2} \tag{5}$$

3 SLAM

3.1 Mapping

The mapping module was optimized by converting data structures to NumPy arrays to enable faster computation.

Cone positions observed multiple times are refined using averaging to improve accuracy. Additional parameters such as threshold distance and averaging factor were introduced to allow easier tuning.

3.2 Localization

Noise was introduced into velocity, heading, and steering inputs to simulate realistic sensor uncertainty.

A simple slip factor was also introduced to model wheel slip that can occur in real-world driving conditions.

3.3 Data Association

The data association module matches newly observed cones with previously mapped cones.

A gating threshold was introduced so that measurements beyond a certain distance are ignored. This prevents incorrect associations and improves the stability of the generated map.

4 Simulation Development

Simulation development was not attempted due to limited experience with the simulation frameworks used in autonomous vehicle environments.

5 Conclusion

This project integrates perception, planning, control, and SLAM modules to create a basic autonomous racing pipeline. The perception system detects cones and estimates their distance, the planner generates waypoints along the track centerline, the controller follows these waypoints using steering and throttle control, and the SLAM system improves localization and mapping accuracy.